

NATIONAL ECONOMICS UNIVERSITY



Customer Churn Prediction and Campaign Analytics for Music Subscription Platforms

Course: Data-Driven Marketing

Group 6

Name	Student ID
Doan Tung Lam	11230555
Nguyen Manh Cuong	112305xx
Tran Thu Hien	11230534
Do Ha Linh	11230557
Pham Minh Bao Ngoc	112305xx

Class: DSEB 65B

Supervisor: Dr. Nguyen Tuan Long

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Abstract

This report presents a complete machine learning pipeline for predicting customer churn on the KKBox music streaming platform, using the WSDM Cup 2018 competition dataset. The pipeline covers raw data preprocessing, temporal feature engineering, model training with hyperparameter tuning, and a business-driven retention layer that converts predicted churn probabilities into per-customer voucher decisions. Feature engineering follows Bryan Gregory’s first-place methodology, producing 53 behavioral and transactional signals. A LightGBM–XGBoost weighted ensemble achieves a validation AUC of 0.769. The retention framework identifies 154,415 high-priority customers for a March 2017 campaign with an estimated ROI of 33–108% depending on budget allocation. A/B testing design and dashboard prototypes are outlined as planned future work.

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1 Introduction

1.1 Business Context

Customer churn—the voluntary cancellation of a subscription—is one of the most costly challenges for streaming platforms. KKBox, a music subscription service based in Taiwan, faces this challenge at scale: acquiring a new subscriber costs significantly more than retaining an existing one, and each churned user represents a lost revenue stream proportional to their remaining expected lifetime.

This project uses data from the WSDM Cup 2018 Kaggle competition, which provides historical transaction records, member demographics, and listening activity logs from KKBox. The task is to predict, at the end of each month, which users will not renew their membership within 30 days of expiry.

1.2 Objectives

The project pursues two parallel goals:

- **Predictive:** Build a model that accurately ranks users by churn probability, evaluated by AUC on the February 2017 validation set and benchmarked against the competition leaderboard.
- **Prescriptive:** Convert churn probabilities into actionable retention decisions—which customers to contact, with what voucher, and in what priority order—grounded in Customer Lifetime Value and individual break-even economics.

1.3 Competition Benchmark

Bryan Gregory's Winning Approach

Bryan Gregory won the WSDM Cup 2018 with a final LogLoss of 0.07974. His key contributions are: (1) the *relative refactoring method* for temporal features—converting dates to days elapsed since the prediction period start so features are comparable across train, validation, and test; (2) iterative cross-validation feature selection from 208 candidates down to 76; and (3) a weighted ensemble of XGBoost (88%) and LightGBM (12%). Over 80% of his features are derived from user activity logs and transaction records.

Our implementation replicates his feature engineering approach on the public dataset and achieves a validation AUC of 0.769. The primary remaining gap is caused by a column name discrepancy in the log data (`num_uniq` vs `num_uniq_songs`), which results in all song-based features being zero in the current run.

1.4 Report Structure

The report proceeds as follows. Section 2 covers data understanding and preprocessing. Section 3 describes temporal feature engineering and the label construction methodology. Section 4 presents the modeling approach and results. Section 5 details the retention decision framework and campaign design. Section 6 outlines the A/B testing plan. Section 7 concludes with findings and future work.

2 Data Understanding & Preprocessing

2.1 Data Sources

The dataset consists of five raw files covering transactions, member profiles, and listening activity. Table 1 summarizes each file.

Table 1: Raw Data Files

File	Rows	Description
transactions.parquet	547,746	Subscriptions, Jan 2015–Jan 2017
transactions_v2.csv	1,431,009	Additional transactions, Jan–Mar 2017
members_v3.csv	6,769,473	User demographics and registration info
user_logs.parquet	106,543	Historical daily listening (22,443 users)
user_logs_v2.parquet	396,362	March 2017 listening (316,345 users)

Key characteristic: Dates are stored as YYYYMMDD integers (e.g., 20170131 represents January 31, 2017) and must be parsed before use. The two transaction files cover overlapping periods and are merged, with duplicates removed.

2.2 Preprocessing Steps

2.2.1 Transactions

The two transaction files are concatenated into a single table of 1,978,751 rows. The following cleaning rules are applied:

- Rows with null `transaction_date` or `membership_expire_date` are dropped.
- For cancellation rows (`is_cancel = 1`, `payment_plan_days = 0`), `actual_amount_paid` is set to zero, since cancelled transactions carry no payment.
- An expiry gap (`expire_gap_days`) is computed as the difference between expiry date and transaction date. Values exceeding 365 days are flagged as extreme (`is_extreme_expiry = 1`).

2.2.2 Members

The member table contains demographic information that requires cleaning:

- **bd** (birth year / age): values outside $[10, 80]$ are replaced with the median, and a binary **bd_missing** flag is created.
- **city, gender, registered_via**: missing values are filled with the string “Unknown” and corresponding missing flags (**city_missing**, etc.) are added.
- **registration_init_time**: decomposed into **registration_year, registration_month, registration_day** and then dropped.

2.2.3 User Activity Logs

The two log files are kept as separate branches rather than merged. The overlap between them is only 4,081 users (18% of the historical log, 1.3% of the March log), confirming that they represent different user cohorts rather than a continuous history.

Log Coverage Limitation

Only 1.1% of the January 2017 training population (10,708 / 992,931 users) appears in the historical log file. This is a known limitation of the public dataset release — KKBox’s internal data includes logs for all users. Zero values in log features are retained as valid signals: a user with no listening history is more likely to churn than an active listener.

2.3 Data Quality Summary

After preprocessing, the clean data contains:

- **Transactions**: 1,978,751 rows, 1,315,711 unique users, no null dates.
- **Members**: 6,769,473 records, 12 columns, no null values in core fields.
- **Logs (historical)**: 106,543 rows, 22,443 users, date range Jan 2015 – Feb 2017.
- **Logs (March)**: 396,362 rows, 316,345 users, date range Mar 2017 only.

3 Feature Engineering

3.1 Data Split and Churn Labels

The time-based split follows Bryan Gregory’s exact configuration to ensure leakage-free evaluation. Table 2 summarizes the three splits.

Table 2: Train / Validation / Inference Split

Split	Month	Users	Churn rate	Label source
Train	Jan 2017	992,931	6.39%	<code>train.csv</code> (official)
Validation	Feb 2017	970,960	8.99%	<code>train_v2.csv</code> (official)
Inference	Mar 2017	907,471	—	<code>sample_submission_v2.csv</code>

Churn labels are taken from the competition’s official files rather than reconstructed internally. This guarantees alignment with competition evaluation.

3.1.1 Label Definition

A user is labelled `is_churn = 1` if they do not renew within 30 days of the `membership_expire_date` of their most recent effective subscription. “Effective” means the state reconstructed at the cutoff using the Scala-aligned sort:

```
sort: transaction_date ASC, is_cancel ASC, membership_expire_date ASC  
  
groupby(msno).last() ⇒ effective state per user at cutoff
```

This logic replicates the official `WSDMChurnLabeller.scala`: the latest transaction wins; on the same date, renewals are preferred over cancellations; among the same type, the longer expiry is taken.

3.2 Anti-Leakage Constraints

All features are computed strictly on data with `transaction_date ≤ cutoff_date`. The cutoff is the last day of the prediction month (e.g., 2017-01-31 for training). The renewal search for label construction is the sole exception: finding a transaction after `last_expire` is part of label logic, not feature logic.

3.3 Temporal Feature Engineering Methods

Bryan Gregory [?] identifies two complementary approaches:

Relative Refactoring Method

A date field is converted to the number of days elapsed since the start of the prediction period. This makes features comparable across train, validation, and test despite covering different calendar months.

Example: `days_since_last_txn = cutoff_date - last_transaction_date`

Absolute Method

The raw date is retained as a YYYYMMDD integer. Used when the calendar date carries signal that elapsed time does not—for example, if users who registered after a holiday promotion have distinct churn behavior.

Example: `registration_date_abs = 20170101`

3.4 Feature Groups

A total of **53 features** are engineered across three source groups.

3.4.1 Member Features (10)

Demographic features (`bd`, `city`, `gender`, `registered_via`), binary missing flags for each, `days_since_reg` (relative method), and `registration_date_abs` (absolute method).

3.4.2 Transaction Features (30)

Key features and their reference to Bryan Gregory’s published top-10 are shown in Table 3.

Table 3: Key Transaction Features

Feature	Bryan ref.	Description
<code>last_plan_days</code>	#1 (6.61%)	Most recent subscription plan length
<code>cancel_to_plandays_ratio</code>	#6 (3.81%)	Cancel count / total plan days
<code>active_no_cancel_flag</code>	#9 (2.80%)	>1 transaction last month, no cancel
<code>last_is_cancel</code>	Paper	Last transaction was a cancellation
<code>cancel_in_last_month</code>	Paper	Any cancellation in last 30 days
<code>avg_cancel_per_month</code>	Paper	Historical cancellations per month
<code>days_since_last_cancel</code>	Paper	Days since most recent cancellation
<code>recency_to_plan_ratio</code>	Higher-order	<code>days_since_last_txn</code> / <code>avg_plan_days</code>
<code>spend_rate_per_day</code>	Higher-order	<code>total_amount_paid</code> / <code>tenure_days</code>

3.4.3 Log Features (13)

User activity features across multiple time windows:

- All-history: `total_secs_played`, `total_uniq_songs`, `num_logins`
- 14-day window (Bryan #2): `secs_played_14d`, `logins_14d`
- 30-day window: `secs_played_30d`, `logins_30d`, `logins_90d`
- Prior-prior month (Bryan #5): `secs_played_60_30d`
- Trend deltas (Bryan #2, #4): `delta_secs_30d_vs_prior`, `delta_songs_14d_vs_30d`
- Recency: `days_since_last_login`, `days_since_significant_usage`
- Binary engagement flag: `has_log_history`

3.5 Feature Summary

Table 4: Feature Engineering Summary

Source	Features	Bryan coverage
Member features	10	Demographic + registration date
Transaction features	30	Covers 7 of Bryan’s Top 10
Log features	13	Covers 5 of Bryan’s Top 10
Total	53	vs 76 in Bryan’s final model

The output `master_model_table.parquet` contains 1,963,891 rows (train + validation) with 53 feature columns and 6 meta columns. Zero NaN values and no meta columns in the feature matrix.

4 Modeling

4.1 Setup and Preprocessing

All models use the same 53-feature matrix. Before training:

- Categorical columns (`city`, `gender`, `registered_via`) are ordinal-encoded using `OrdinalEncoder` with unknown values mapped to `-1`.
- Numeric columns are imputed with training-set medians (applied to validation and inference without look-ahead).
- **For Logistic Regression only:** features are standardized with `StandardScaler`.
- Class imbalance (14.6:1 non-churn to churn) is handled via `class_weight='balanced'` for Logistic Regression, `is_unbalance=True` for LightGBM, and `scale_pos_weight=14.64` for XGBoost.

4.2 Models

4.2.1 Logistic Regression (Baseline)

Logistic Regression provides a linear lower bound. It confirms that the feature set carries meaningful signal: an AUC of 0.702 means features such as `auto_renew_rate`, `days_since_reg`, and `registered_via` have strong monotone relationships with churn.

Settings: `solver='saga'`, `C=1.0` (L2), `max_iter=1000`.

4.2.2 LightGBM

LightGBM uses leaf-wise tree growth—each split maximizes impurity reduction for a single leaf, producing deeper trees with fewer nodes than level-wise approaches. This is well-suited to the sparse log features and high-cardinality categoricals.

Known Limitation

The current run shows `best_iteration = 6`, meaning early stopping fires after only 6 boosting rounds. This is caused by all song-based log features being zero (the `num_uniq` column name fix was applied after this run). A re-run after the fix is expected to produce a proper convergence curve and reduce LogLoss from 0.298 toward the `<0.15` range.

Hyperparameter tuning: Optuna TPE sampler, 100 trials, objective = maximize val AUC. Searched: `num_leaves` $\in [20, 300]$, `learning_rate` $\in [0.001, 0.3]$, `feature_fraction`, `bagging_fraction`, L1/L2 regularization.

4.2.3 XGBoost

XGBoost uses level-wise tree growth with `tree_method='hist'` (Bryan’s explicit recommendation for speed at high dimensionality). This is more stable than leaf-wise but slower.

XGBoost achieves higher AUC (0.768 vs 0.761) but worse LogLoss (0.545 vs 0.298). The AUC advantage means XGBoost ranks users better in absolute terms, but its probability estimates are less calibrated. Since the competition metric is LogLoss, LightGBM is the stronger single model on this run.

4.3 Results

Table 5: Model Performance on Validation Set (February 2017)

Model	Val AUC	Val LogLoss	Notes
Logistic Regression	0.702	0.607	Linear baseline
LightGBM	0.761	0.298	Best LogLoss
XGBoost	0.768	0.545	Best single AUC
Ensemble (95/5)	0.769	—	Grid-searched blend
<i>Bryan Gregory (1st place): LogLoss = 0.07974</i>			
<i>Bryan no-logs baseline (17 features): LogLoss = 0.10453</i>			

4.4 Ensemble Construction

The final prediction is a weighted average:

$$\hat{p}_{\text{churn}} = w_{\text{LGBM}} \cdot \hat{p}_{\text{LGBM}} + w_{\text{XGB}} \cdot \hat{p}_{\text{XGB}} \quad (1)$$

Bryan Gregory used $w_{\text{LGBM}} = 0.12$, $w_{\text{XGB}} = 0.88$, derived from 84 leaderboard submissions. A grid search over $w_{\text{LGBM}} \in [0, 1]$ in 5% steps yields an optimal blend of 95/5 on this run, because our LightGBM has better LogLoss calibration. This diverges from Bryan’s weights because his XGBoost was the dominant model (76 features, full log data), whereas our LightGBM outperforms XGBoost here.

Raw probabilities are further calibrated using isotonic regression fitted on the validation set. The calibrated mean prediction (8.7%) closely matches the observed February churn rate (8.99%).

4.5 Gap Analysis

Table 6: Gap Analysis: Our Model vs Bryan Gregory

Dimension	Our model	Bryan Gregory
LogLoss (best)	0.298 (LGBM)	0.07974
AUC (ensemble)	0.769	≈ 0.97
Features	53	76
Log user coverage	1.1%	Full (internal)
LGBM best iteration	6 (degenerate)	Converged
Ensemble weights	95/5 LGBM/XGB	12/88 LGBM/XGB

The 0.22 LogLoss gap decomposes into three causes: (1) the `num_uniq` column bug zeroing all song features; (2) the resulting LGBM degenerate early stopping (`best_iter = 6`); and (3) the 98.9% zero coverage of log features in the public dataset. Bryan’s no-logs baseline of 0.10453 with only 17 features shows that, with proper feature engineering and full model convergence, a score in the 0.08–0.10 range is achievable on this data.

5 Retention Decision

The retention layer converts continuous churn probabilities into binary intervention decisions. The guiding principle: a voucher should only be sent when the expected economic benefit of retention exceeds the cost of the incentive.

5.1 Behavioral Weight Computation

Rather than assuming fixed multipliers, behavioral adjustments to the save rate are computed directly from training data. For each binary behavior flag i :

$$\text{relative_effect}_i = \frac{\text{churn_rate}(i=1) - \text{churn_rate}(i=0)}{\text{overall_churn_rate}} \quad (2)$$

$$w_i = \text{clip}(\text{relative_effect}_i \times 0.25, -0.25, +0.25) \quad (3)$$

A positive weight means the behavior is associated with lower churn (increases save rate); a negative weight means higher churn (decreases save rate).

Table 7: Data-Driven Behavioral Weights (computed from training data)

Behavior flag	Definition	Churn (=0)	Churn (=1)	Weight
Auto-renew active	<code>last_is_auto_renew=1</code>	8.9%	5.1%	+0.125
Recent cancellation	<code>last_is_cancel=1</code>	7.6%	14.6%	−0.225
Has log history	<code>has_log_history=1</code>	7.7%	8.3%	−0.021
Active, no cancel	<code>active_no_cancel=1</code>	7.5%	62.1%	−0.250
High listening vol.	<code>total_secs > Q75</code>	7.7%	8.3%	−0.021
Long plan user	<code>avg_plan_days ≥ 90</code>	7.4%	72.4%	−0.250

Counterintuitive Findings

Two results in Table 7 contradict common assumptions:

Long-plan users churn at 72.4%. Users who purchase 90-day plans but still churn are likely one-time promotional buyers, not committed subscribers. The long commitment signals an initial promotional purchase rather than genuine loyalty.

Active, no-cancel users churn at 62.1%. This flag captures users who have recent transactions and no cancellation history—but they are approaching natural membership lapse without actively cancelling. They are “passive churners” on the way out, not engaged subscribers.

These patterns are only discoverable through data-driven weight computation.

5.2 Quantile-Based Risk Tiers

Fixed thresholds (e.g., 0.30, 0.50, 0.70) become meaningless when the model output is narrow or bimodal. Quantile-based boundaries adapt to the actual distribution:

Table 8: Risk Tier Configuration (Q1=0.030, Q2=0.038, Q3=0.074)

Tier	P_churn range	Voucher	Base save rate	Interpretation
Stable	$< Q_1$	0%	0%	Likely to renew; no action
Medium	$[Q_1, Q_2)$	5%	8%	Undecided; moderate response
High	$[Q_2, Q_3)$	10%	15%	Sweet spot; highest responsiveness
Critical	$\geq Q_3$	20%	7%	Disengaged; harder to recover

The adjusted save rate per user is:

$$\text{save_rate}_i = \text{clip}\left(\text{base_save_rate} \times \left(1 + \sum_j w_j \cdot \mathbf{1}[\text{flag}_{ij} = 1]\right), 0.05, 0.65\right) \quad (4)$$

5.3 Customer Lifetime Value

CLV estimates the forward-looking revenue at risk if the customer churns today:

$$\text{monthly_revenue}_i = \frac{\text{avg_amount_paid}_i}{\text{avg_plan_days}_i / 30} \quad (5)$$

$$\text{expected_lifetime} = \frac{1}{\text{validation_churn_rate}} = \frac{1}{0.0899} \approx 11.1 \text{ months} \quad (6)$$

$$\text{CLV}_i = \text{monthly_revenue}_i \times \text{expected_lifetime} \quad (7)$$

The validation churn rate is read from the training data, not hardcoded, so CLV updates automatically when the model is retrained. The resulting CLV distribution has a median of \$1,434 and a range of \$0–\$21,881.

5.4 Retention Cost and Break-Even

$$\text{retention_cost}_i = \min(\text{avg_amount_paid}_i \times \text{voucher_pct}, \$200) \quad (8)$$

The \$200 cap prevents over-spending on premium users. The per-customer break-even churn probability is:

$$p_i^* = \frac{\text{retention_cost}_i}{\text{CLV}_i \times \text{save_rate}_i} \quad (9)$$

A voucher is sent only when *all four* conditions hold:

1. Risk tier is not Stable.
2. $p_{\text{churn},i} > p_i^*$ (per-customer break-even satisfied).
3. $p_{\text{churn},i} \geq t^*$ (global profit-optimal floor; see §5.5).
4. $\mathbb{E}[\text{profit}_i] = p_{\text{churn},i} \cdot \text{CLV}_i \cdot \text{save_rate}_i - \text{retention_cost}_i > 0$.

5.5 Threshold Sweep

A global floor t^* is swept from the 10th to 95th percentile of non-Stable eligible users. At each threshold:

$$\text{net_profit}(t) = \sum_{i: p_i \geq t} (p_i \cdot \text{CLV}_i \cdot \text{save_rate}_i - \text{cost}_i) \quad (10)$$

Table 9: Threshold Sweep — Selected Decision Points

Threshold	Users	Cost	Net profit	ROI
0.058	149,201	\$2,840,230	\$939,748	33.1%
0.096	47,730	\$1,533,949	\$692,895	45.2%
0.393	12,035	\$406,486	\$437,237	107.6%
Selected (t^*)	154,415	\$2,891,910	\$954,847	33.0%

Table 9 reveals a clear budget-ROI tradeoff: contacting all economically justified users maximizes absolute net profit (\$955K), while tightening to $t = 0.393$ contacts only 12K users but nearly doubles the return on each dollar spent (108% ROI).

5.6 Strategic Segmentation and Campaign Design

5.6.1 Segment Definitions

Customers are classified into four strategic segments based on CLV and churn risk:

Table 10: Strategic Segments and Retention Outcomes

Segment	CLV	Tier	Users	Net benefit
VIP	High	Critical	46,457	\$669,090
High Persuadable	High	High	56,299	\$156,752
Critical Selective	Any	Critical	1,274	\$23,814
Targeted	Any	Med/High	50,385	\$105,191
Stable – No Action	Any	Stable	188,460	\$0
Do Not Target	—	—	564,596	—

5.6.2 Recommended Action per Segment

- **VIP:** Premium rescue offer with personalized message and 20% voucher (capped at \$200). Highest business priority.
- **High Persuadable:** Standard 10% voucher via in-app notification. Most users in this group are still engaged and respond well to a timely offer.
- **Critical Selective:** Capped voucher only for users where $\mathbb{E}[\text{profit}] > 0$. Do not over-spend on users unlikely to respond.
- **Targeted:** Standard tier-based voucher (5–10%).
- **Stable – No Action:** Monitor renewal behavior; no spend required.

5.6.3 Budget Recommendation

For cost-constrained campaigns, running **VIP + Critical Selective only** is the recommended starting point:

Users contacted	47,731
Total cost	\$1,534,000
Net benefit	\$693,000
ROI	45%

This captures 73% of the total net benefit at 54% of the full campaign cost.

5.6.4 Timing and Delivery

Vouchers are delivered 7–10 days before membership expiry, when the renewal decision is most salient. Delivery channels: in-app notification (primary) and email (secondary). Vouchers are personalized with the user’s listening history where available.

The output `campaign_tracking_template.csv` contains placeholder fields for post-campaign measurement: `voucher_redeemed`, `renewed`, `actual_churn`, and `actual_profit`. These are populated after the campaign runs, enabling direct comparison of predicted vs. actual ROI.

6 A/B testing

Status: Planned – Not Yet Executed

This section describes the intended experimental design. The A/B test has not been run yet. Results, statistical analysis, and dashboard visualizations will be added upon campaign completion.

6.1 Objectives

The experiment answers two causal questions that the predictive model cannot answer alone:

1. Does sending a retention voucher actually increase renewal rates among targeted users, and by how much?
2. Is the tier-based variable voucher size more effective than a flat discount?

Without randomized assignment, any observed difference could reflect selection bias (we targeted users who would have renewed anyway), so the control group is essential to establish the true incremental lift.

6.2 Experimental Design

6.2.1 Population and Randomization

The 154,415 targeted users are randomly assigned at the individual user level (not device or session) to prevent contamination. Randomization is stratified by risk tier (Stable / Medium / High / Critical) to ensure balanced tier representation in each arm.

Table 11: Experimental Arms

Arm	Allocation	Users	Treatment
Control	20%	30,883	No voucher, standard renewal flow
Treatment A	40%	61,766	Tier-based voucher (5 / 10 / 20%)
Treatment B	40%	61,766	Flat 10% voucher for all tiers

6.2.2 Primary Metric and Hypothesis

The primary metric is the 30-day renewal rate:

$$\text{lift} = \frac{r_{\text{treatment}} - r_{\text{control}}}{r_{\text{control}}} \quad (11)$$

Null hypothesis: $H_0 : r_{\text{treatment}} = r_{\text{control}}$. Alternative hypothesis: $H_1 : r_{\text{treatment}} > r_{\text{control}}$ (one-sided, $\alpha = 0.05$).

6.2.3 Sample Size and Statistical Power

Using the February 2017 baseline renewal rate of 91.01% (100% – 8.99% churn rate), targeting a minimum detectable effect of $\delta = 2$ percentage points at 80% power and $\alpha = 0.05$:

$$n = \frac{(z_\alpha + z_\beta)^2 \cdot (p_0(1 - p_0) + p_1(1 - p_1))}{(p_1 - p_0)^2} \approx 12,000 \text{ users per arm} \quad (12)$$

The planned allocation of 61,766 per treatment arm exceeds this threshold, providing sufficient power to detect smaller effects.

6.3 Secondary Metrics

- Revenue per user in the 30-day post-expiry window.
- Voucher redemption rate per tier (Treatment A only).
- Actual ROI vs model-predicted ROI per segment.
- False-positive cost: voucher cost incurred on users who would have renewed without intervention.

6.4 Analysis Plan

1. **Primary comparison:** Two-proportion z-test comparing renewal rates between Treatment A vs Control and Treatment B vs Control.
2. **Heterogeneous effects:** Two-way ANOVA by risk tier and CLV quartile to determine whether the voucher effect varies across customer segments.
3. **Multiple comparisons:** p-values adjusted using the Benjamini–Hochberg procedure (controlling false discovery rate at 5%).
4. **Economic validation:** Compare actual campaign ROI (from `campaign_tracking_template.csv` filled post-campaign) against the model-predicted ROI of 33–108%.

6.5 Dashboard Prototypes (Planned)

Five dashboards are planned to monitor and report results. Figures will be inserted upon completion.

6.5.1 Executive KPI Dashboard

Audience: business leadership. Key metrics: active subscriber count, monthly churn rate trend, campaign ROI to date, predicted vs actual churn comparison. Updated monthly.

[Figure: Executive KPI Dashboard — placeholder]

6.5.2 Model Performance Dashboard

Audience: data science team. AUC and LogLoss on rolling validation windows, P_churn distribution histogram, feature importance top 20, prediction calibration curve (predicted vs actual churn rate by decile).

[Figure: Model Performance Dashboard — placeholder]

6.5.3 Segment Performance Dashboard

Audience: product and marketing. Churn rate, CLV, and renewal rate by strategic segment (VIP / High Persuadable / Critical Selective / Targeted / Stable). Enables segment-level trend monitoring across months.

[Figure: Segment Performance Dashboard — placeholder]

6.5.4 Campaign Dashboard

Audience: marketing operations. Live campaign metrics: vouchers sent, redemption rate by tier, renewal rate by arm, cost vs net benefit running total. Linked to `campaign_tracking_template.csv`.

[Figure: Campaign Dashboard — placeholder]

6.5.5 A/B Testing Dashboard

Audience: analytics team. Real-time experiment progress: arm allocation and balance, running renewal rate per arm, lift estimate with 95% confidence interval, p-value trend, and projected statistical significance date.

[Figure: A/B Testing Dashboard — placeholder]

7 Conclusion

7.1 Summary of Findings

This project delivers a complete churn prediction and retention optimization pipeline for a music streaming platform. The key contributions are:

1. **Temporal feature engineering:** 53 features built using Bryan Gregory’s relative refactoring and absolute temporal methods, covering 8 of his 10 published top features. The pipeline is fully leakage-proof, with all features computed strictly on data available at the prediction cutoff.
2. **Ensemble model:** A LightGBM–XGBoost weighted blend achieves validation AUC = 0.769 and LogLoss = 0.298. The primary gap to the competition benchmark (0.07974) is caused by a column name bug in the log features (`num_uniq` vs `num_uniq_songs`) and is expected to close significantly upon re-run.

3. **Data-driven retention economics:** Individual break-even thresholds replace a universal decision rule. Each user's voucher decision is grounded in their own CLV, behavioral save-rate adjustment, and retention cost. The result is a campaign targeting 154,415 users with an estimated ROI of 33–108% depending on budget.
4. **Behavioral insights:** Data-driven weight computation reveals that long-plan users churn at 72% and active-no-cancel users at 62%—both counterintuitive patterns that assumed multipliers would have gotten wrong. These findings have direct implications for product design (long-plan promotions may be attracting low-loyalty customers) and campaign targeting (active-no-cancel users are passive churners, not loyal ones).

7.2 Limitations

- **Log coverage:** 98.9% of training users have zero log features in the public dataset. Song-based features (`total_uniq_songs`, `delta_songs_14d_vs_30d`) are currently zero due to the column name bug.
- **Model calibration:** LightGBM's `best_iteration = 6` means the model barely learned. Predicted probabilities cluster narrowly around 3–7%, limiting tier differentiation.
- **CLV assumptions:** CLV uses the February 2017 churn rate for expected lifetime. This is a single-period estimate; a survival analysis approach would be more rigorous.
- **Save rate estimates:** The base save rates (7–15%) are calibrated to the training data behavioral churn differences. Actual campaign save rates may differ and should be updated from A/B test results.

7.3 Next Steps

1. Re-run Feature Engineering after the `num_uniq` fix. Expected outcome: all 53 features populated, LGBM converges properly, LogLoss drops from 0.298 toward <0.15.
2. Increase Optuna trials to 100+ (already configured). This is expected to further reduce LogLoss by 0.01–0.03.
3. Execute the A/B test described in Section 6 and populate `campaign_tracking_template.csv` with actual outcomes.
4. Build the five planned dashboards and replace placeholders in Section 6.
5. Explore survival analysis (Cox proportional hazards) as a more principled CLV model for heterogeneous user lifetimes.